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## Rendering Overview

When you have finished creating a composition in Fusion, you need to render the files out to disk for playback and integration into a larger timeline. Fusion Studio and the Fusion page in DaVinci Resolve use very different workflows for rendering. To finish a composite in the Fusion Page, you use a MediaOut node to cache the results into the Edit or Cut page Timeline. The DaVinci Resolve Deliver page handles the final render of the entire Timeline. To get completed composites out of Fusion Studio, you configure and render them starting with a Saver node in the Node Editor. Fusion Studio is also capable of distributing a variety of rendering tasks to other machines on a network.

### Rendering in the Fusion Page

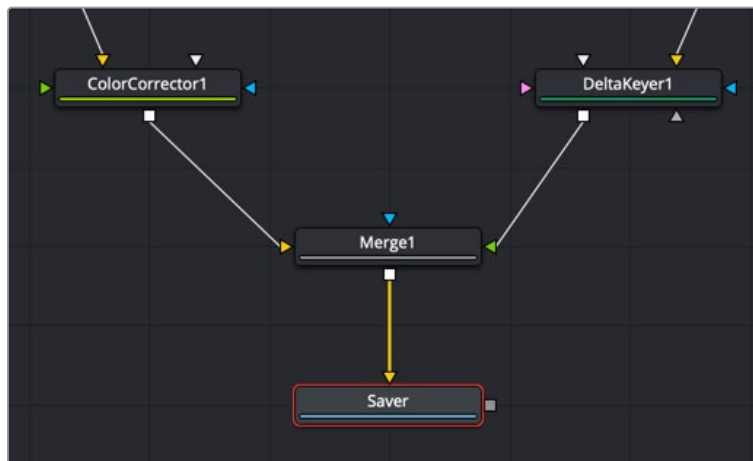
In the Fusion page, a MediaOut node is required for getting your composite from the Fusion page back into the Edit or Cut page Timeline. Whatever the MediaOut node displays when you see it in the viewer is what gets rendered back into the Edit or Cut page. This process is semi-automatic in DaVinci Resolve, where the Smart Render Cache setting begins caching the MediaOut node almost immediately when you return to the Edit or Cut page Timeline. The cache file format and any resolution scaling to fit the composition into the Timeline Resolution is handled in the DaVinci Resolve Project Settings.

### Rendering in Fusion Studio

In Fusion Studio, all rendering goes through Saver nodes. Similar to MediaOut nodes in the Fusion page, Saver nodes are most often appended to the end of a node tree to render the final composite. The Saver node determines the name, format, and location of the rendered files.

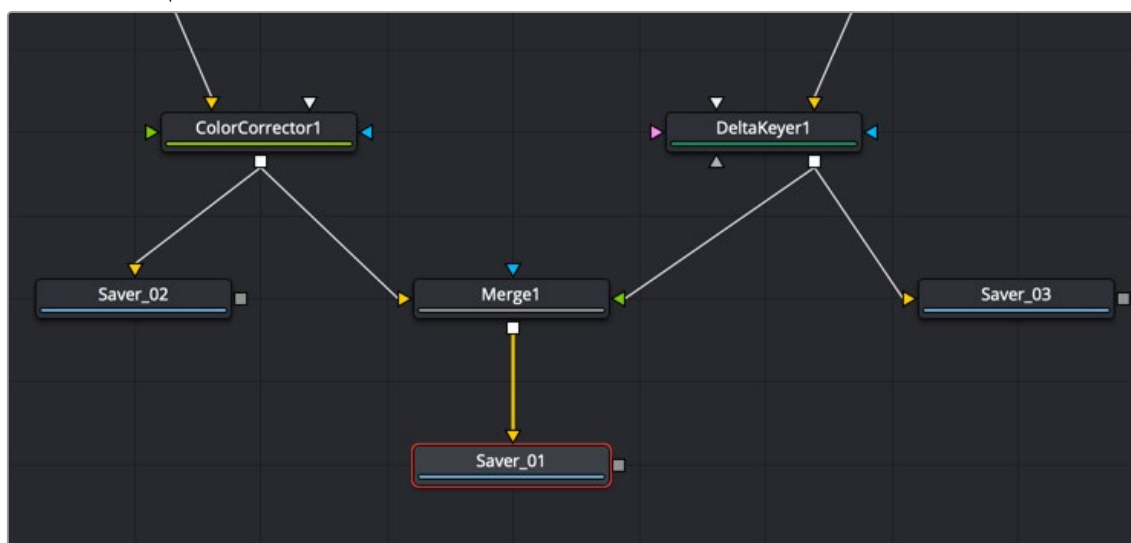
# Rendering with the Saver Node

To begin rendering in Fusion Studio, you must add at least one Saver node to the node tree. Most of the time, you will place at least one Saver node at the very end of your tree to render the final image.



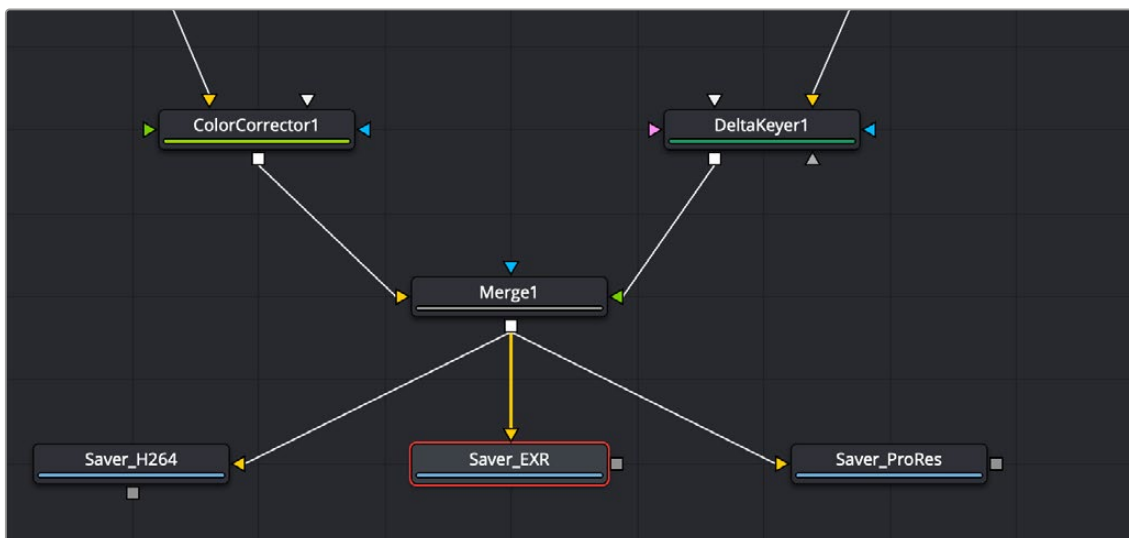
A single Saver node is added to the end of a node tree to render the final composite.

You can attach multiple Saver nodes anywhere along the node tree to render out different parts of a composite. In the example below, three Saver nodes are added at different points in the node tree. The top two render out each half of the composite while the bottom renders the results of the entire composite.



Multiple Saver nodes can be added to different parts of a node tree.

You can also use multiple Saver nodes stemming from the same node in order to create several output formats. The below example uses the three Savers to export different formats of the same shot.



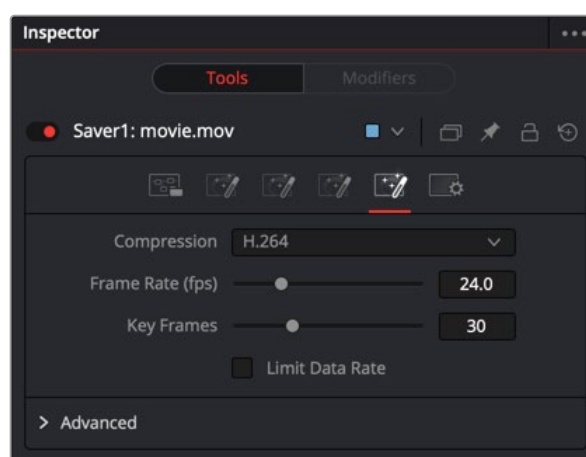
Multiple Saver nodes can be added to create different formats for output.

Adding a Saver node to a node tree automatically opens a Save dialog where you name the file and navigate to where the exported file is saved. You can then use the Inspector to configure the output format.

For more information on the Saver node, see *Chapter 105, “I/O Nodes,”* in the DaVinci Resolve Reference Manual or Chapter 45 in the Fusion Reference Manual.

## Setting Filenames for Export

If you use a file extension when naming the file, Fusion will set the output format accordingly. For example, naming your file `image_name.exr` will set the Inspector to output an EXR file or naming a file `image_name.mov` will set the Inspector for an H264 QuickTime movie. If you decide to change or modify the setting of the file type, the Saver’s format tab in the Inspector contains the specific parameters for the selected format.



The Saver’s Format tab with controls for QuickTime

If you decide to output an image sequence, a four-digit frame number is automatically added before the filename extension. For example, naming your file `image_name.exr` results in files named `image_name0000.exr`, `image_name0001.exr`, and so on. You can specify the frame padding by adding several zeroes to indicate the number of digits. For example, entering a filename as `image_name_000.exr` results in a sequence of images with the names `Image_name_000.exr`, `Image_name_001.exr`, `Image_name_002.exr`, and so on.

**NOTE:** The starting frame number always uses the Time Ruler start frame number.

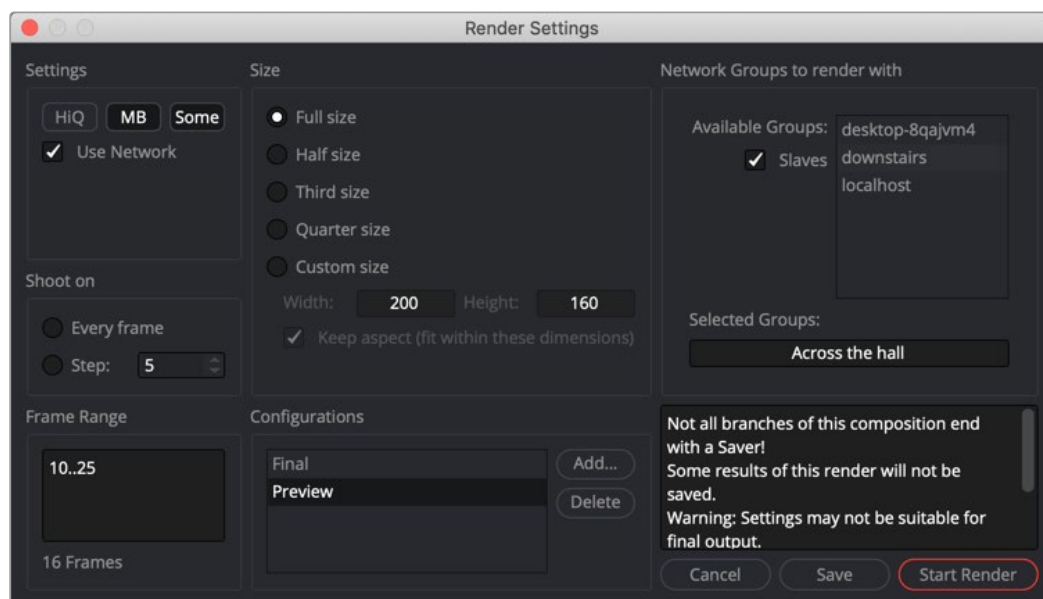
## Using the Render Settings Dialog

In Fusion Studio, you initiate rendering by clicking the Render button and opening the Render Settings dialog. This dialog configures the quality, frame range, and network usage of the rendering.

To Render a comp in Fusion Studio:

- 1 Connect a Saver node at the end of your composition.
- 2 Enter a name and a location for the saved file(s) in the Save window.
- 3 Set the format using the Format tab in the Inspector, if necessary.
- 4 Click the Render button in the transport controls area or choose File > Render All Savers.

The Render Settings dialog opens providing options for the rendered output.



The Render Settings dialog options

Ensure that the frame range and other parameters are correct and click Start Render.

## Saver nodes in the DaVinci Resolve Fusion page

The Fusion page also includes a Saver node, although it is used for a different purpose than the Saver node in Fusion Studio. Rendering from the Fusion page is handled primarily through the MediaOut node. There is no Render Settings dialog since the rendering location and format is predetermined by DaVinci Resolve's cache settings. Saver nodes in the Fusion page are a special case, and are used to render OpenEXR files only. Once you add a Saver node, you enter the filename with the extension. `exr`. Click the Browse button to select a location for the EXR sequence, and then choose Fusion > Render All Savers.

Using the Saver node is useful for optimizing extremely complex and processor-intensive compositions. For example, you can render out specific branches of a node tree that no longer requires frequent adjustment to OpenEXR via a Saver node, and then reimport the result to take the place of the original branch of nodes in order to improve the performance of your composition.

Alternatively, you can render out multi-channel mattes or EXR images containing Arbitrary Output Variables (AOVs) to bring into other applications.

## Render Settings Dialog Options

Many of the options in the Fusion Studio Render Settings dialog are used when you need to create a quick preview or a test render. The options in this dialog allow you to increase performance by disabling some of the image-processing operations that are time consuming but deliver higher quality results. Often the first settings you set are those in the Configurations section. This section determines whether you want to produce a final high quality render or a faster preview render. Selecting Final prevents you from modifying the options that will limit the quality.

### Settings

When the Configuration section is set to Preview, the Settings section of the Render dialog includes three options that determine the overall quality and appearance of your final output. These buttons also have a significant impact on render times. When the Configurations setting is set to Final, these options cannot be disabled

**HiQ:** When enabled, this setting renders in full image quality. If you need to see what the final output of a node would look like, then you would enable the HiQ setting. If you are producing a rough preview to test animation, you can save yourself time by disabling this setting.

**MB:** The MB in this setting stands for Motion Blur. When enabled, this setting renders with motion blur applied if any node is set to produce motion blur. If you are generating a rough preview and you aren't concerned with the motion blur for animated elements, then you can save yourself time by disabling this setting.

**Some:** When Some is enabled, only the nodes specifically needed to produce the image of the node you're previewing are rendered.

## Size

When the Configurations section is set to Preview, you can use the Size options to render out frame sizes lower than full resolution. This is helpful when using the Render dialog to create proxies or just creating a smaller file size.

## Network

The Network setting controls the distribution of rendering to multiple computers. For more information, see the network rendering section in this chapter.

## Shoot On

Again, this option is only available when Configurations is set to Preview. The Shoot On setting allows you to skip frames when rendering. You can choose to render every second, third, or fourth to save render time and get faster feedback. You can use the Step parameter to determine the interval at which frames are rendered.

## Frame Range

Regardless of whether the Configurations is set to Final or Preview, this option defaults to the current Render In/Out Range set in the Time Ruler to determine the start and end frames for rendering. You can modify the range to render more or fewer frames.

## Configurations

When set to Final, the Render Settings are set to deliver the highest quality results, and you cannot modify most of the options in this dialog. When set to Preview, you can set the options to gain faster rendering performance. Once you've created a useful preview configuration, you can save it for later use by clicking the Add button, giving it a name, and clicking OK.

## Rendering Previews

You can render Flipbook previews into a viewer. These Flipbook previews exist entirely within RAM. They are created by right-clicking over a node in the Node Editor and choosing Create > Play/Preview on > Left viewer/Right viewer from the drop-down menu. The Render Settings dialog appears where you can configure the preview and initiate the rendering. You can also Option-drag a node directly from the Node Editor into a viewer. The Render Settings dialog will be displayed, and the preview will appear on the viewer you target.

For more information on rendering RAM previews, see *Chapter 69, "Using Viewers."* in the DaVinci Resolve Reference Manual or Chapter 7 in the Fusion Reference Manual.

**TIP:** Option-Shift-dragging a node into a viewer will skip the Render dialog and previously used settings.

# Setting Up Network Rendering in Fusion Studio

Fusion Studio is capable of distributing a variety of rendering tasks to an unlimited number of computers on a network, allowing multiple computers to assist with creating network-rendered previews, disk caches, and final renders.

Using the Render Settings dialog or the built-in Render Manager, you can submit compositions to be rendered by other copies of Fusion Studio, as well as to one or more Fusion Render nodes. Rendering can also be controlled through the command line for integration with third-party render managers like Deadline, Rush, and Smedge.

Render nodes are computers that do not have the full Fusion application installed but do have Fusion Render node software installed. The Render node software is not installed by default when you install Fusion Studio, but it can be installed at any time using the Fusion Render node Installer. The installer is located in the Blackmagic Fusion Studio installer.dmg on macOS and the Blackmagic Fusion Studio.zip on Linux and Windows. Fusion Studio is licensed for an unlimited number of Render nodes, so you can install the Render node software on as many macOS, Windows, and Linux computers that you want involved in network rendering.

## To install a Render node:

- 1 Download and unzip the Blackmagic Fusion Studio archive. Then locate the Render Node Installer.
- 2 Copy the Install Fusion Render Node [version] to each computer on the network that you want to perform rendering operations.
- 3 Install the Render node.

By default, the Render node application will be added to the Start Menu on Windows under Blackmagic Design. On macOS, it is added to the menu bar, and on Linux it appears in the app launcher. Each time you log in to the computer, the Render node application will run automatically.

To disable the Render node application from starting up automatically, choose Quit from the Render node icon in the macOS menu. On Linux, right-click over the icon and choose Kill Process, and on Windows, delete the shortcut from the Windows Startup directory.

## Licensing for Network Rendering

Most versions of Fusion Studio are licensed by connecting a single-seat hardware key (dongle) to the same computer where Fusion is installed. Each dongle includes an unlimited number of cross-platform Render node licenses, which you can install on as many macOS, Windows, and Linux computers as you need. For Fusion Studio to access the Render nodes, the computer with the Fusion Studio dongle needs to be on the same local network subnet as the Render nodes. The network licensing does not require individual license files; instead, the Render nodes automatically search for the dongle on the subnet, making it easy to set up. Single-seat dongles do not “float” over a network; they must be connected to the same computer where Fusion Studio operates.

## Multi-License Dongles

Using a multi-license dongle, you can license 10 copies of Fusion Studio by connecting the dongle to any computer on the same subnet. Since these licenses “float” over a network, Fusion Studio does not have to be running on the same computer where the dongle is connected. As long as Fusion Studio is on the same subnet, it can automatically find the license server and check out an available license.

Multi-seat dongles can be combined together to tailor the number of Fusion seats in a larger facility. For example, three dongles each licensed for 10 Fusion Studios would serve up 30 licenses. This also allows for redundancy. For instance, in the example above, three computers can act as license servers. If the first server fails for some reason, Fusion Studio will automatically try the next server. Alternatively, multiple dongles can also be plugged into a single computer.

## Setting Up a License Server

Setting up the license for network rendering begins by connecting either a single-seat or multi-seat dongle to a computer that will act as the host for the license server. The Render node installer installs Fusion Server, which acts as the license server, although the Render node does not have to run on that same computer. The Fusion Server is set up to launch at startup and run unobtrusively in the background as a service/daemon, ready to serve licenses and Fusion bins. The Fusion Server is used to serve up multiple licenses so it must be running whenever you want to operate Fusion Studio or use the Render nodes for network rendering. Its default is to start up automatically and remain running as long as a license is being used by another computer or a Render node is operating. If nothing is using the Fusion Server, it will quit after 30 seconds.

You will need your network administrator to set firewall rules allowing the Fusion Server, FusionScript, and the Fusion Render node applications to communicate and confirm licensing with the computer that has the Fusion Studio dongle.

If for some reason you remove a dongle or the network drops out, the licenses of any connected Fusion Studio application will also drop. Upon losing its license, Fusion Studio will start searching for another license, locally or on a different machine. If no license is found, Fusion pauses rendering and displays a dialog with options to retry the search or autosave the comp and quit. Render nodes only check for a license on the network once during startup, so they are not affected by removing the dongle or network issues.

## Setting Up a License Server with Environment Variables

Environment variables provide a way to specify flexible or “variable” configuration options. When network rendering with Fusion, environment variables can be useful for temporarily setting a location or choosing a preference file. Using the `FUSION_LICENSE_SERVER` environment variable, you can set different locations for the File Server.

Instead of looking in a single location for the Fusion Server, you can set up multiple license servers separated by semicolons. For instance, `fu:SetPrefs("Global.EnvironmentVars.FUSION_LICENSE_SERVER", "192.168.1.12; 192.168.10.55;*")`

You can also use the environment variable to scan for license servers within a subnet—for example, `"bobs-mac.local;10.0.0.23;*;license.mystudio.com"`. Including an asterisk (\*) indicates a broadcast search of the local subnet.

Like most environment variables, you can put the license server in the Global Preferences via the Prefs text file. EnvironmentVars: `fu:SetPrefs("Global.EnvironmentVars.FUSION_LICENSE_SERVER", "10.0.0.23;*") fu:SavePrefs()`, see *Chapter 75, “Preferences,”* in the DaVinci Resolve Reference Manual or *Chapter 15* in the Fusion Reference Manual for more information on using environment variables.

**NOTE:** The use of straight quotes (") in the environment variables above are intentional and should not be replaced with typographer’s, or curly, quotes (“”).



## Configuring the Render Master and Render nodes

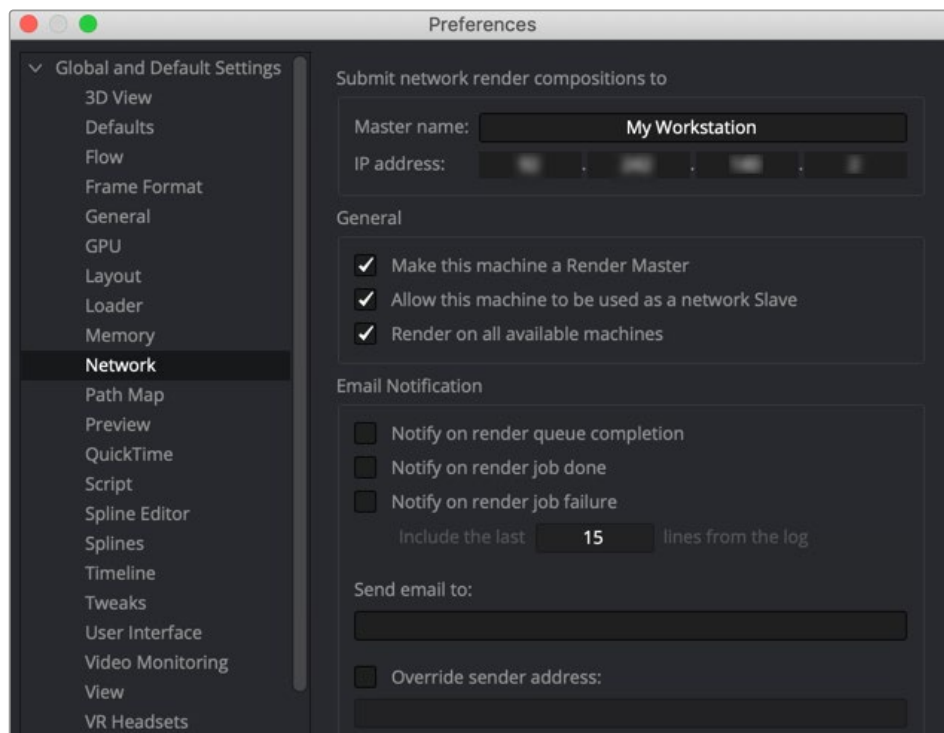
There are two roles played by computers involved in network rendering.

- The Render Master manages the list of compositions to be rendered (the queue) and allocates frames to Render nodes for rendering. Metaphorically speaking, the Render Master is the traffic cop of this process.
- The Render nodes are the main computers used for the rendering process. All computers involved in network rendering must be on the same network subnet, and they all must have access to the various files (including Fonts and third-party plugins) used to create the composite. The path to the files must be the same for each computer involved in rendering.

### Preparing the Render Master

The Render Master manages the list of compositions to be rendered (the queue) and allocates frames to Render nodes for rendering. The Render Master is also used to maintain the Render node list and to push updates to the various Render nodes when needed. At least one computer in the render farm must be configured to act as the Render Master.

Any copy of Fusion can act as a Render Master by setting up the Fusion Network Preferences.



The Network panel sets up computers as Render Masters.

Acting as a Render Master has no significant impact on render performance. The system resources consumed are insignificant. However, there are specific steps you must take to make one of your computers a Render Master.

To set up the Render Master:

- 1 Install a copy of Fusion Studio on the computer you want to be the Render Master.
- 2 In Fusion Studio, choose Fusion Studio > Preferences on macOS or File > Preferences on Windows and Linux
- 3 In the Preferences dialog, select the Global > Network Preferences panel.

- 4 Enter the name of the Render Master in the Name field and enter the IP address.
- 5 Enable the Make This Machine a Render Master checkbox.
- 6 If you want to use this computer as part of the render farm, enable the Allow This Machine to Be Used as a Render Node checkbox as well.

**To have the Render Node act as the Render Manager:**

- Select a node in the Render Manager and choose Set Default Master from the contextual menu.

Once a computer is enabled to act as the master, use the Render Manager to add the Render nodes it will manage. The Render Manager dialog is described in detail later in this chapter.

## Preparing Render Nodes

Before you can begin rendering on the network, the Render nodes must be set up to accept instructions from the Render Master.

**In Fusion Studio, you can enable the computer to be used as a Render node in two ways:**

- Choose File > Allow Network Renders.
- Enable the Allow This Machine to Be Used as a Network Render Node in the Global > Network Preferences.

**On a macOS Render Node computer:**

- Click the Render Node icon in the menu bar and choose Allow Network Renders.



The Render Node menu  
accessed in the macOS menu bar

**On a Linux Render Node computer:**

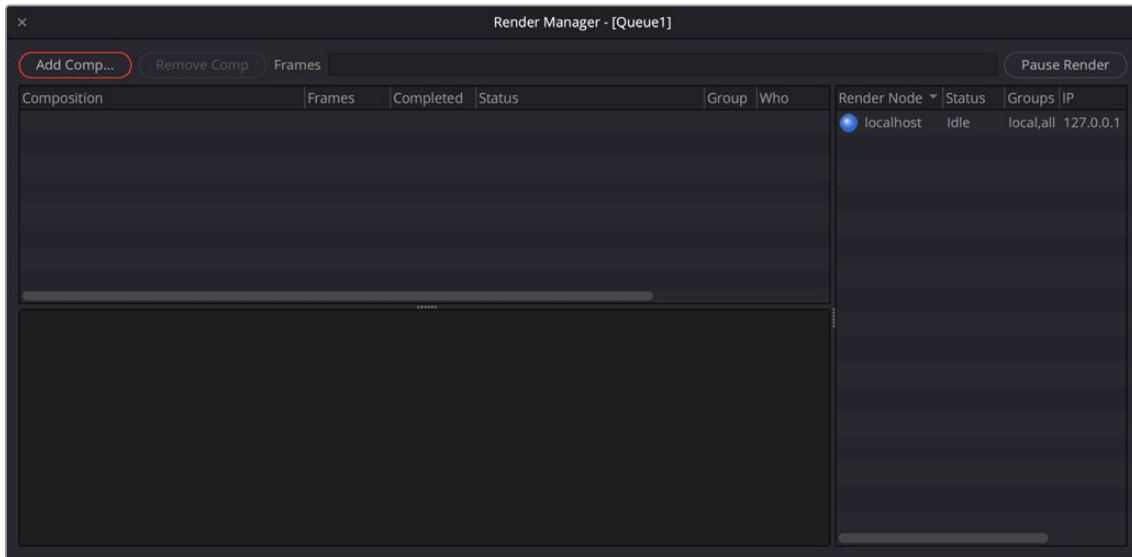
- Right-click the Fusion Render Node icon in the App Launcher and choose Allow Network Renders.

**On a Windows Render Node computer:**

- Right-click the Fusion Render Node icon in the taskbar Notification area and choose Allow Network Renders.

## Setting Up the Render Manager

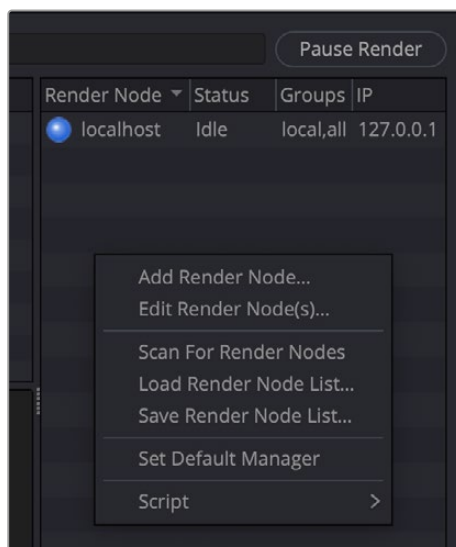
The Render Manager window is used to monitor the progress of rendering. It can be used to reorder, add, or remove compositions from a queue, and to manage the list of Render nodes used for rendering. To open the Render Manager window in Fusion Studio, choose File > Render Manager.



The Render Manager is used to reorder, add, and remove compositions from a render queue.

The Render Master is always listed as the first computer in the Render Node list along the right side. This allows the Render Manager to render local queues without using the network. For the Render Master to control additional Render nodes, the nodes must be added to the Render Node list.

Right-clicking in the Render Node list allows you to add Render nodes by entering the Render node's name or IP address. You can also choose Scan to have the Render Manager look for Render nodes on the local network.



The Render Manager is used to add Render nodes.

## Scanning for Render Nodes

With the Render Manager open, you can scan for Render nodes by right-clicking in the Render Manager's Render Node list and choosing Scan for Render Nodes from the drop-down menu.

Scanning looks through all IP addresses on the network subnet to determine whether any other computers in the local network are actively responding on the port Fusion uses for network rendering. A copy of the Fusion Render node must be running on the remote computer in order for it to be detected by the scan.

## Manually Adding Render Nodes

To manually add a Render node to the Render Node list, right-click in the Render Manager's Render Node list and choose Add Render Node from the drop-down menu.

The screenshot shows a dark-themed dialog box titled "Render Node". It contains the following elements:

- Render Node name:** A text input field with a red border.
- IP address:** Four numeric input fields, each containing the number "0".
- Available Groups:** A list box containing the items "all" and "local".
- Selected Groups:** A list box containing the item "all".
- Buttons:** "Cancel" and "OK" buttons at the bottom right, with the "OK" button highlighted by a red border.

The Add Render Node dialog allows you to manually enter a Render Node name or IP address to locate it on the subnet.

In the Add Render Node dialog that opens, enter the name or the IP address of the remote Render node. The Render Manager will attempt to resolve names into IP addresses and IP addresses into names automatically. You can use this method to add Render nodes to the list when they are not currently available on the network.

## Removing Render Nodes

To remove a computer from the Render Node list, right-click over the Render Node in the Render Node list and choose Remove Render Node(s) from the pop-up menu. You can use Command on macOS or Ctrl on Windows and Linux to select multiple Render nodes for removal.

## Loading and Saving Render Node Lists

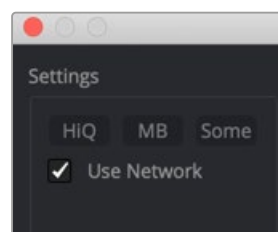
The list of Render nodes is automatically saved in the Documents > Blackmagic Design > Fusion > Queue folder when you quit the Render Manager. You can save and reload alternative lists of Render nodes by choosing Render Node > Save Render Node List and Load Render Node list from the menu.

# Submitting Comps to Network Render

To submit a comp to render on the network, you can use the Render Manager, the Render Settings dialog, or a third-party render farm application. The Render Settings dialog is quicker, while the Render Manager and third-party render farm applications can provide more feedback and control over the process.

## Using the Render Settings Dialog for Network Rendering

When starting a preview or a final render, selecting the Use network checkbox from the Render Settings dialog and submitting the render adds a composition to the end of the current queue in the Render Manager. The Render Master used is based on the Fusion preferences from the workstation submitting the comp.



The Use Network checkbox enables network rendering from the Render Settings dialog.

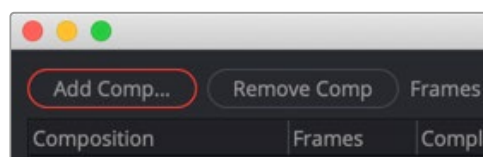
**NOTE:** Distributed network rendering works for image sequences like EXR, TIFF, and DPX. You cannot use network rendering for Quicktime, H264, ProRes, or MXF files.

## Using the Render Manager Window for Network Rendering

The Render Manager uses a render queue that lets you batch render comps. Compositions are rendered in the order in which they are listed in the Render Manager, with the top entry rendered first, followed by the next item down, and so on. Multiple comps in a queue may render simultaneously depending on the group of Render nodes they are using and the priority assigned to each comp.

### To add a comp to the queue in the Render Manager:

- Click the Add Comp button and navigate to the comp on your hard drive.
- Right-click in the queue list and select Add Comp from the drop-down menu, and then navigate to the comp on your hard drive.
- Drag a comp file from an OS window into the Render Manager's queue list.



The Add Comp button in the Render Manager adds a comp to the queue for batch rendering over the network.

## Removing a Composition from the Queue

To remove a composition from the queue, select the composition in the queue list and press the Backspace/Delete key or right-click over the comp in the queue list and choose Remove Composition from the drop-down menu.

## Saving and Reloading Queue Lists

It can be useful to save a queue list to reuse at a later time. The current queue list is saved in the Documents > Blackmagic Design > Fusion > Queue folder. To save the current queue with a new name, choose File > Save Queue As in the Render Manager menu bar. To reload a saved Queue, choose File > Load Queue and navigate to the saved location.

## Reordering the Queue Lists

In the middle of a job, priorities for finishing a composition may change. Shifting deadlines may require that a composition further down in the queue be rendered sooner. You can move comps to a new position in the queue by dragging them in the queue list. If a composition with a status set to Done is moved lower in the queue, it does not re-render. To re-render a comp, right-click on the comp in the queue and choose Clear Completed frames from the drop-down menu.

## Working with Node Groups

Render nodes can be configured into Groups, which are then used when submitting compositions. For example, imagine you have five Render nodes. All the Render nodes are members of the group named All. Two of the Render nodes include more memory and faster processors, so you create a new group called Hi-Performance.

New Render nodes are automatically added to All, but you can assign them to other groups as well.

### To assign a Render node to a Group:

- 1 Open the Render Manager and select the Render nodes to assign to a group.
- 2 Choose Render Node > Assign Group or right-click over the Render nodes and choose Assign Group from the drop-down menu.
- 3 In the Choose Group dialog, enter a name for the group.
- 4 To assign Render nodes to multiple groups, separate the name of each group using a comma (e.g., All, Local, or Hi-Performance). The order of the groups determines the priority. See “Using Multiple Groups” below.

When a render is submitted to the network, it is automatically sent to the All group. However, you can choose to submit it to other groups in the list.

### To submit a comp to a group from the Render Manager:

- 1 Open the Render Manager.
- 2 Submit the comp.
- 3 Click the Pause Render button.
- 4 Right-click over the comp in the queue list and select Assign Group.
- 5 In the Assign Group dialog, select an existing group to render the comp and click OK.
- 6 Click the Resume Render button.

**To submit a comp to a group from the Render Settings dialog:**

- 1 Click the Render button in the transport controls area.
- 2 In the Render Settings dialog, enable the Use Network checkbox.
- 3 Select an existing group from the Available groups list.
- 4 Click the Start Render button.

Continuing with the group example above, five Render nodes are contained in the All group, and two of those Render nodes are also in the Hi-Performance group. If you submit a render to the Hi-Performance group, only two of the computers on the network are used for rendering. If a composition is then submitted to the All group, the remaining three machines will start rendering the new composition. Once the two Render nodes in the Hi\_Performance group complete the first job, they join the render in progress on the All group.

Groups are optional and do not have to be used. However, groups can make managing large networks of Render nodes easier and more efficient.

**Using Multiple Node Groups**

A single Render node can be a member of multiple groups. A single composition can also be submitted to multiple groups. Submitting a composition to multiple groups results in it rendering on all Render nodes in the selected groups.

When a Render node is a member of multiple groups, the order of the groups is important because the order defines the priority for that Render node.

For example, if groups are assigned to a Render node as All, Hi-Performance, then renders submitted to the All group take priority. Any renders in progress that were submitted to the Hi-Performance group will be overridden. If the order is changed to Hi-Performance, All, then the priority is reversed.

## Viewing the Render Log

The Render Log is displayed in the lower half of the Render Manager window, although it can also be displayed in the console window. The text in the log displays the Render Manager activities, including which frame is assigned to which Render node, which Render nodes have loaded the compositions in the queue, and statistics for each render after completion.

**To view the Render Log in the console:**

- Open the Render Manager and choose Misc > Show Render Log.
- There are two modes for the Render Log: a Verbose mode and a Brief mode. Verbose mode logs all events from the Render Manager, while Brief mode logs only which frames are assigned to each Render node and when they are completed.

**To disable Verbose mode:**

- Choose Misc > Verbose Logging from the Render Manager's menu bar.

## Using Third-Party Render Managers with Fusion Studio

You can make use of third-party render manager software to control network rendering. This allows for efficient sharing of your computer resources between the many applications that may make use of them. Examples of such managers are Smedge from Uberware LLC, Rush from Seriss, and Deadline from GetRender. Generally, these render managers use a command line renderer. By default, Fusion's Render nodes operate as a service to the Fusion internal Render Manager. However, you can also run the Render nodes via the command line for third-party render managers.

Keep in mind that using a third-party render manager will prevent the use of some of Fusion's network rendering features, such as the ability to create network rendered Flipbook Previews and disk caches.

### Command Line Rendering

For studios using third-party render farm managers like Deadline, Smedge, or Rush, the Fusion Render node can be called via command line passing arguments and file paths. In this Windows example, a Render node is called to load a composition called exampleV001, and render 10 frames:

```
//pathtoRN/FusionRenderNode.exe //pathtoProject/exampleV001.comp -render -start 101 -end 110 -quit
```

This would start up, render frames from 101 to 110, and then quit.

The following table lists additional command line features.

| Command                        | Description                                                                                                                   |
|--------------------------------|-------------------------------------------------------------------------------------------------------------------------------|
| "Fusion Server -i"             | Installs the license server as a service or daemon, launching it on startup before user login.                                |
| "Fusion Server -S" (capital S) | Causes the Fusion Server to run persistently in the background, until force-quit.                                             |
| <filename.comp>                | Full path and comp name, like /storage/project/episode/shot/filename.comp.                                                    |
| -render                        | Tells the Render Node to render.                                                                                              |
| -frames <frameset>             | Passes a series of frame ranges to be rendered—e.g., 101..110,120,121,130..150.                                               |
| -start <frame>                 | Sets the start frame of the render.                                                                                           |
| -end <frame>                   | Sets the last frame of the render.                                                                                            |
| -step <step>                   | Normally set to 1, step skip frames for rendering. For instance, 2 would render every second frame                            |
| -quit                          | Causes the Render Node to quit after the render is complete.                                                                  |
| -join <host>                   | Prompts the node to connect to a manager at <hostname,IP>, and (re)join any ongoing renders.                                  |
| -listen                        | The node remains running and waits for incoming requests from a manager.                                                      |
| -log <filename>                | Causes the Render Node to output information about the render to a log file. This appends to the end of an existing log file. |
| -cleanlog                      | Clears existing text from a log file.                                                                                         |
| -verbose                       | Outputs more detailed information into the log file.                                                                          |



| Command                           | Description                                                                                                                                     |
|-----------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| -quiet                            | Suppresses pop-ups and interface buttons from displaying and needing interaction.                                                               |
| -version                          | Returns the Render Node version number.                                                                                                         |
| -pri highlabove normal below idle | Sets the node's process priority to high, above, normal, below, or idle.                                                                        |
| -args <arg1> [, <arg2> ...]       | Allows storing custom values that can be fetched by calling the script function GetArgs(), which will return a table of { <arg1>, <arg2>, ... } |

**TIP:** An X11 virtual frame buffer is required to make a headless Linux command line interface work.

## Preparing Compositions for Network Rendering

The way you construct a composition in Fusion Studio can help or hinder network rendering. The media you read in, where plugins are installed, and the mix of operating systems on your networked computers all play a part in how smoothly your network rendering goes. Your setup must include several essential parts before network rendering will work:

- License dongle, Render Master, and Render nodes must be on the same local network (subnet).
- Fusion Server must be running as a background service on the same computer where the dongle is installed.
- All source media from the comp should be placed on a network volume.
- The network volume must be mounted on each Render node.
- Loaders must point to the media on the mounted volumes.
- Savers must write to a drive that is mounted on each Render node.
- The Fusion comp must be saved to a volume that is mounted on each Render node.
- All Render Nodes and Render Masters need read and write access to any volumes specified as a source media location or render destination.
- Make sure all fonts used in the comp for Text+ and 3D text nodes are installed on all the Render nodes
- Make sure all Render nodes have third-party OFX plugins installed if any are used in the comp.

Below are more details about some of these items.

## Using Relative Paths

The file paths used to load a composition and its media, and to save the composition's rendered results, are critical to the operation of network rendering. Each computer used for network rendering must be able to access the media location for each Loader in the comp. Savers must be set up to save to folders that all Render nodes can see and to which all Render nodes have write permission. Even the composition must be saved in a folder accessible to all Render nodes, and it should be added to the queue list using a path that's visible to all Render nodes.

For example, if you open a composition located at `c:\compositions\test1.comp` in Fusion Studio and add the composition to the network rendering queue, the Render Manager sends a message to each Render node to load the composition and render it. The problem is that each computer is likely to have its own `c:\` drive that does not contain the comp you created. In most cases, the Render nodes will be unable to load the composition, causing it to fail.

Path Maps located in Fusion Preferences are virtual paths used to replace segments of file paths. They can change the absolute paths used by Loader and Saver nodes to relative paths. There are a number of Path Maps already in Fusion, but you can also create your own. The most common path to use is the `Comp:\` path.

`Comp:\` is a shortcut for the folder where the actual composition is saved. So, using `Comp:\` in a Loader makes the path to the media file relative, based on the saved location of the comp file. As long as all your source media is stored in the same folder or subfolder as your comp file, Fusion locates the media regardless of the actual hard drive name.

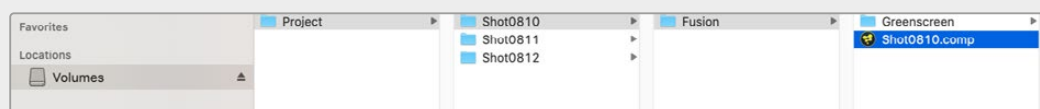
Here's an example of a file structure that enables you to use relative file references. The composition is stored in the following file path:

```
Volumes\Project\Shot0810\Fusion\Shot0810.comp
```

And your source media is stored here:

```
Volumes\Project\Shot0810\Fusion\Greenscreen\0810Green_0000.exr
```

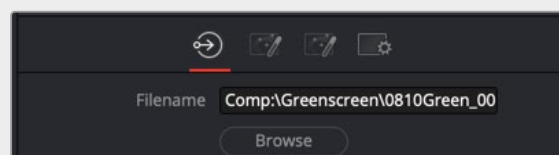
This overall directory structure can be seen in the following screenshot:



File paths can use relative paths based on the location of the saved comp file.

In this situation, using the `Comp:\` path means your media location starts from your comp file's location. The relative path set in the Loader node would then be:

```
Comp:\Greenscreen\0810Green_0000.exr
```



Replacing the Loader's path to start with `Comp:\` creates a relative path from the comp file's location.

If your source media's actual file path uses a subfolder in the same folder as the comp file's folder:

```
Volumes\Project\Shot0810\Footage\Greenscreen\0810Green_0000.exr
```

The relative path set in the Loader node would then be:

```
Comp:..\Footage\Greenscreen\0810Green_0000.exr
```

The two dots .. instruct the path to go up one folder.

**TIP:** Some Path Maps are not set up on a Fusion Render Node automatically. For instance, you must manually add an entry for macros if you are using macros in your comp.

## Using Mapped Drives

Having the comp and media set to relative paths solves only part of the problem. Each computer involved in rendering needs to have access to the drive where the comp and source media are located. Mapping a drive provides permanent access to a folder on another computer or network storage device.

Windows assigns a new drive letter to the folder, and it can be accessed just like any other drive connected to your computer. Mapped drives assign a letter of the alphabet to a shared network resource. Your shared drives must be the same drive letters on all Render nodes. For example, if your media is on drive Z, then the network drive must appear as the letter Z on each of the Render nodes.

On macOS, you can map a network drive using Connect to Server from the Go menu. Entering the smb:// path to the drive will mount it on the computer. Using Accounts > Login Items, you can have the network drive auto-mount after a reboot as well.

## Installing All Fonts on Render Nodes

All fonts used by Text tools in the composition must be available to all nodes participating in the render. The render will otherwise fail on the Render nodes that do not have the font installed.

## Installing Third-Party Plugins on Render Nodes

All third-party plugins and tools used by a composition must be installed in the plugins directory of each Render node. A Render node attempting to render a composition that uses a plugin that's not installed will fail to render. Licensed plugins are required on each Render node.

## Other Uses of Network Rendering

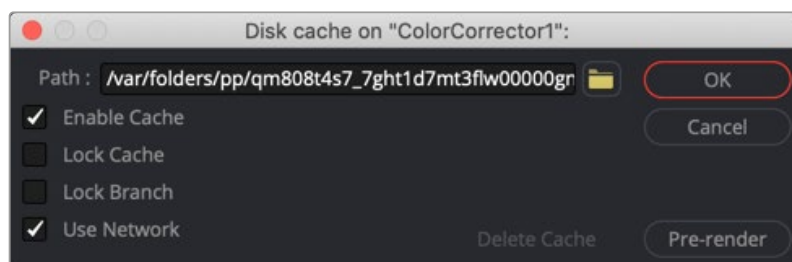
Although you will probably set up network rendering for the purpose of accelerating the output of your final renders, Fusion is capable of using the network for other purposes as well. You can use the Render nodes on your network to accelerate the creation of Flipbook Previews and disk caching as well.

### Flipbook Previews

Fusion Studio is able to use Render nodes to accelerate the production of Flipbook Previews, allowing for lightning fast previews. Frames for previews that are not network rendered are rendered directly into memory. Select the Use Network checkbox and the Render nodes to render the preview frames to the folder set in the Preferences Global > Path > Preview Renders. This folder should be accessible for all Render nodes that will participate in the network render. The default value is Temp\, which is a virtual path pointing to the system's default temp folder. This will need to be changed before network rendered previews can function. Once the preview render is completed, the frames that are produced by each Render node are spooled into memory on the local workstation. As each frame is copied into memory, it is deleted from disk.

### Disk Cache

Right-clicking a node in the Node Editor and choosing Cache to Disk opens a dialog used to create the disk cache. If you enabled the Use Network checkbox and click the Pre-Render button to submit the disk cache, the network Render nodes are used to accelerate the creation of the disk cache.



Render nodes can be used for disk caching as well as final renders.

## When Renders Fail

It is a fact of life that render queues occasionally fail. The composition has an error, the power goes out, or a computer is accidentally disconnected from the network are some causes for failure. If no one is available to monitor the render, the risk that an entire queue may sit inactive for several hours may become a serious problem.

Fusion Studio includes a variety of measures to protect the queue and ensure that the render continues even under some of the worst conditions.

### Automatic Rejoining of the Queue

If a Render node becomes unavailable to the Render Master for any reason, frames assigned to that Render node are reassigned among the remaining Render nodes in the list.

When the Render node becomes available for rendering again, it will signal the Render Master that it is ready to render again, and new frames will be assigned to that Render node.

This is why it is important to set the Render Master in the network preferences of the Render nodes. If the master is not set, the Render node will not know what master to contact when it becomes available.

In the Fusion Render Node Preferences, select the Tweaks panel. Using the Last Render Node Restart Timeout field, you can enter the number of seconds Fusion waits after the last Render node goes offline before aborting that queue and waiting for direct intervention.

## Relaunching Render Nodes with Fusion Server

Fusion Server is a small utility installed with Fusion Studio and the Render node. The application is Fusion Server is a small utility installed with Fusion Studio and the Render node. The application is silently launched by each Fusion Render node when started.

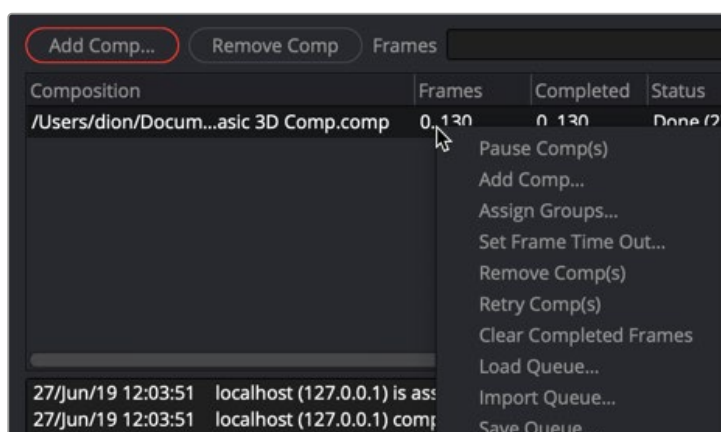
Fusion Server monitors the Render node to ensure that the Render node is still running during a render. It consumes almost no CPU cycles and very little RAM. If the monitored Render node disappears from the system's process list without issuing a proper shutdown signal, as can happen after a crash, the Fusion Server relaunches the Render node, allowing it to rejoin the render.

Fusion Server will only detect situations where the Render node has exited abnormally. If the Render node is still in the process list but has become unresponsive for some reason, the Fusion Server cannot detect the problem. Hung processes like this are detected and handled by frame timeouts, as described below.

## Frame Timeouts

Frame timeouts are a fail-safe method of canceling a Render node's render if a frame takes longer than the specified time (with a default of 60 minutes, or one hour). The frame timeout ensures that an overnight render will continue if a composition hangs or begins swapping excessively and fails to complete its assigned frame.

The timeout is set per composition in the queue. To change the timeout value for a composition from the default of 60 minutes, right-click on the composition in the Render Manager's queue list and select Set Frame Timeout from the contextual menu.



Right-click over a comp in the Render Manager to set a timeout value.

To change the frame timeout value, choose Set Frame Time Out from the Render Manager's Misc menu and enter the number of seconds you want for the Time Out.

## Heartbeats

The Render Master regularly sends out heartbeat signals to each node, awaiting the node's reply. A heartbeat is basically a message from the manager to the Render node asking if the node is still responsive and healthy. If the Render node fails to respond to several consecutive heartbeats, Fusion will assume the Render node is no longer available. The frames assigned to that Render node will be reassigned to other Render nodes in the list.

The number of heartbeats in a row that must be missed before a Render node is removed from the list by the manager, as well as the interval of time between heartbeats, can be configured in the Network Preferences panel of the master. The default settings for these options are fine for 90% of cases.

If the compositions that are rendered tend to use more memory than is physically installed, this will cause swapping of memory to disk. It may be preferable to increase these two settings somewhat to compensate for the sluggish response time until more RAM can be added to the Render node.

## Managing Memory Use

Often, the network environment is made up of computers with a variety of CPU and memory configurations. The memory settings used on the workstation that created a composition may not be appropriate for all the Render nodes in the network. The Render node software offers the ability to override the memory settings stored in the composition and use custom settings more suited to the system configuration of a specific Render node.

To access preferences for a node, right-click on the icon in the Windows Notification area or from the macOS menu bar and choose Preferences. In the Preferences dialog, select the Memory panel.

### Override Composition Settings

Enable this option to use the Render node's local settings to render any incoming compositions. Disable it to use the default settings that are saved into the composition.

### Render Several Frames at Once

Fusion has the ability to render multiple frames at once for increased render throughput. This slider controls how many frames are rendered simultaneously. The value displayed multiplies the memory usage (a setting of 3 requires three times as much memory as a setting of 1).

Normal values are 2 or 3, although machines with a lot of memory may benefit from higher values, whereas machines with less memory may require the value to be 1.

### Simultaneous Branching

Enable this option to render every layer in parallel. This can offer substantial gains in throughput but may also use considerably more memory, especially if many layers are used in the composition. Machines with limited memory may need to have Simultaneous Branching disabled when rendering compositions with many layers.

# Limitations of Render Nodes

There are a few important limitations to remember while setting up compositions and rendering over a network.

## Time Stretching

Compositions using the Time Stretcher and Time Speed tools may encounter difficulties when rendered over the network. Speeding up or slowing down compositions and clips requires fetching multiple frames before and after the current frame that is being rendered, resulting in increased I/O to the file server. This may worsen bottlenecks over the network and lead to inefficient rendering. If the composition uses the Time Stretcher or Time Speed tools, make certain that the network is up to the load or pre-render that part of the composition before network rendering.

## Linear Tools

Certain tools cannot be network rendered properly. Particle systems from third-party vendors, such as Genarts's Smoke and Rain, and the Fusion Trails node cannot render properly over the network. These tools generally store the previously rendered result and use it as part of the next frame's render, so every frame is dependent on the one rendered before it. This data is local to the tool, so these tools do not render correctly over a network.

## Saving to Multi-Frame Formats

Multiple machines cannot render a single QuickTime file. Always render to separate sequential file formats like EXR, DPX, JPEG, and so on. Once the render is complete, a single workstation can load the image sequence in order and save to the desired compiled format.

**NOTE:** The above does not apply to network rendered previews, which are previews created over the network that employ spooling to allow multi-frame formats to render successfully. Only final renders are affected by this limitation.

# Troubleshooting

There are some common pitfalls when rendering across a network. Virtually all problems with network rendering have to do with path names or plugins. Return to the "Preparing Compositions for Network Rendering" section in this chapter to review some of the essential setup requirements. Verify that all Render nodes can load the compositions and the media, and that all Render nodes have installed the plugins used in the composition.

If some difficulties persist, contact Blackmagic Design's technical support using the support section on the Blackmagic Design website. Save a copy of the render.log file to send to technical support.

## Checking the Render Log

The log file shown in the Render Manager dialog displays messages that can assist with diagnosing why a render or node has failed. The render log shows a step-by-step account of what happened (or didn't happen) during a render. If a Render node cannot be found, fails to load a composition or render a frame, or simply stops responding, it will be recorded here.

## Check the Composition

The Render Manager's Status field in the render log indicates if a composition fails to render. Some possible causes of this are as follows:

- **No Render Nodes Could Be Found:** On the Preferences Network tab, make sure that there is at least one Render node available, running and enabled. If all Render nodes are listed as Offline when they are not, check the network.
- **The Composition Could Not Be Loaded:** Some Render nodes may not be able to load a composition while others can. This could be because the Render node could not find the composition (check that the path name of the composition is valid for that Render node) or because the composition uses plugins that the Render node does not recognize.
- **The Render Nodes Stop Responding:** If a network link fails, or a Render node goes down for some reason, the Render node will be removed from the active list and its frames will be reassigned. If no more Render nodes are available, the composition will fail after a short delay (configurable in network preferences). If this happens, check the render log for clues as to which Render nodes failed and why.
- **The Render Nodes Failed to Render a Frame:** Sometimes a Render node simply cannot render a particular frame. This could be because the Render node could not find all the source frames it needed, or the disk it was saving to become full or because of any other reason for which Fusion might normally be unable to render a frame. In this case, the Render Manager will attempt to reassign that failed frame to a different Render node. If no Render node can render the frame, the render will fail. Try manually rendering that frame on a single machine and observe what happens.
- **Check the Render Nodes:** Fusion's Render Manager incorporates a number of methods to ensure the reliability of network renders. Periodically, the Render Manager will send signals called Heartbeats, generated at regular intervals, to detect network or machine failures. In this event, a failed Render node's outstanding frames are reassigned to other Render nodes where possible. In rare cases, a Render node may fail in a way that the heartbeat continues even though the Render node is no longer processing. If a Render node failed (although the Render Master may not have detected it) and you do not want to wait for the Frame Timeout, simply restart the Fusion workstation or Fusion Render Node that has hung. This triggers the heartbeat check, reassigns the frames on which that Render node was working, and the render should continue. Heartbeats may fail if the system that is performing the render is making extremely heavy use of the Swap file or is spending an extraordinary amount of time waiting for images to become available over a badly lagged network. The solution is to provide the Render node with more RAM, adjust memory settings for that node, or upgrade the network bandwidth.
- **Check the Network:** At the Render Master, bring up the Network tab of the Preferences dialog box and click Scan. If a Render node is not listed as running, the Render Master will not be able to contact it for network rendering. Alternatively, bring up a command prompt and ping the Render nodes manually. If the remote systems do not respond when they are up and running, the network is not functioning and should be examined further.